

The Ethics of AI in Warfare

GLBL 290 Final Group Paper

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War has been nearly ever-present in the history of humanity. Across the globe, wherever remnants of ancient civilizations are discovered, signs of warfare in that society are usually quickly found. However, while the existence of war has been a constant throughout history, its methods and ways of fighting have not. Developments in technology have essentially always been used by militaries to try to gain an advantage over the adversary. The Bronze Age led to the mass development of swords and spears. The chemical discovery of gunpowder became the necessary ingredient in guns and cannons. And the culmination of decades of physics research among the world's most brilliant minds resulted in the atomic bomb.

This trend of technological breakthroughs leading to novel weapons has only accelerated in recent decades. In a world of growing geopolitical tension, the emphasis on developing the strongest and most capable weapons is enormous. Following this historical trend, the most transformative technology of today - Artificial Intelligence - is quickly being used in new weapons. The recent breakthrough in AI research has led to a race to develop powerful weapons infused with AI. But in a world where the next breakthrough in AI is uncertain, and the consequences of war are most dire, it is all the more critical to study the ethical considerations of AI in war. To understand the ethics of AI weapons, this paper reviews a brief history of the laws of war, the usage of AI in war today, what governmental regulation of AI weapons may look like, and the economic impacts of government investments in AI weapons.

Brief History of Laws of War

To properly situate the role of law and ethics in war today, it's helpful to take a quick look at the history of the laws of war. In general, before the 19th century, formal laws of war between fighting militaries were rare. However, there were typically rules of conduct, mostly based on the cultural values of the civilizations at war. For example, in medieval times, under the code of chivalry, it was common for knights to take injured enemy soldiers as prisoners and ransom them for their freedom. Meanwhile, these same knights would typically have no qualms about slaughtering lower-class civilian peasants. The rules of war were highly cultural and varied largely among different militaries.

In the 19th century, a paradigm shift occurred in the theory of war. *On War*, published in 1832 and written by General Carl von Clausewitz, reshaped how people thought about war. Declaring war to not be its own independent act between states but rather “the continuation of politics by other means”, Clausewitz reframed war as an instrument used by politicians rather than merely a battle between opposing generals. This reframe provided a theoretical foundation for the eventual formal laws of war: if war was a political instrument and not an inevitable act between militaries, then just like other interactions between states, it could be governed and restricted with rules.

Throughout the first half of the 20th century, as World Wars I and II unleashed an unprecedented level of death and destruction, a formal set of laws of warfare was established over a set of international treaties. Culminating in the Geneva Convention, the foundation of International

Humanitarian Law (IHL) was established. New laws of war have been added as new weapons have been developed, but the guiding principles of IHL were established in these treaties.

While the details and nuances of IHL are broad, the core ideas of IHL are the following: distinction of combatant vs civilian; proportional attacks to limit civilian casualties; humane treatment for all. AI weapons may prove to be some of the most advanced and lethal weapons in existence, yet if used ethically, they can increase military adherence to IHL and decrease unnecessary civilian suffering in war.

Current Usage of AI Weapons

To develop an understanding of the ethical implications of AI weapons, it's first necessary to examine how these weapons are being used today, along with the role policy choice plays in affecting the usage of advanced weapons. Further, how can these new developments affect the way that IHL can be observed by militaries? To dive into these questions, two research papers were reviewed. The first, 'Double Standards: The Implications of "Near" Certainty Drone Strikes in Pakistan' by Raman, Lushenko, and Kreps, reviewed how a change in official policy on drone strikes significantly lowered the number of civilian casualties by the US in Pakistan. The other article, 'Digital Targeting: Artificial Intelligence, Data, and Military Intelligence' by King, surveyed the usage of AI weapons by the Ukrainian military against Russia. Taken together, these two papers provide the guidelines for how human-in-the-loop (HITL) AI weapons, along with appropriate strategic guidelines, can effectively accomplish military goals while minimizing civilian harm.

The Role of Policy in Adhering to International Law

The paper ‘Double Standards: The Implications of “Near” Certainty Drone Strikes in Pakistan’ provides empirical evidence that a change in policy on the usage of advanced weapons can have a direct impact on decreasing civilian casualties. In their paper, the researchers studied the change in the number of civilian casualties from US drone strikes in Pakistan before and after the implementation of a new directive by President Obama in 2013 meant to reduce civilian casualties. The order instructed the military to only authorize drone strikes when there was a “near” certainty that there would be zero civilian casualties in a given strike.

The researchers found a significant drop in the number of civilian casualties following the implementation of the policy. Prior to President Obama’s official announcement in May of 2013, there was an average of over 9 civilian casualties per month. The number dropped to just over 0.2 following his policy change. Meanwhile, the strike precision - the percentage of people killed in the drone strike who were the intended target - jumped from 76.1% to 97%, a remarkable feat. This reduction in civilian casualties, using weapons with advanced guidance and analytics, shows that new weapon systems have the ability to reduce the collateral damage of ongoing military operations when used along with guidelines meant to protect civilians.

The Role of AI Weapons in the Ukrainian Military

Next, in the paper ‘Digital Targeting: Artificial Intelligence, Data, and Military Intelligence’, the methods of using AI by the Ukrainian military are explored. Using a qualitative method of analysis, the paper surveys the usage of AI weapons against Russia. The key finding is that human-out-of-the-loop (HOTL) systems are not yet ready for combat, but that HITL systems

have been widely used and very effective. The key difference between HITL and HOTL systems is the degree to which humans are involved in approving decisions. In an HITL system, any major decision, such as completing a strike, must be approved by a designated human commander. Meanwhile, HOTL systems are able to, once activated, decide to complete a potentially lethal strike without first receiving direct approval.

HITL systems represent a clear extension of the advanced analytics weapons developed over the previous few decades. The key difference is that in those past systems, large amounts of data, like satellite images, needed to be reviewed by humans before a strike was possible. This was a laborious process that occupied valuable time used by troops. Now, as described in the paper, much of this work can be automated by systems that are trained to quickly and effectively identify military targets using AI. By contrast, HOTL systems represent a break from this method of warfare. Additionally, they are practically much more difficult to implement. While there may be clear military benefits to an HOTL drone that can complete a complex mission without a human having to pilot it, this represents a huge technical challenge not yet possible.

Global Governmental Regulations on AI in Warfare

Another important aspect in examining how AI is being integrated into warfare is how governments around the world are regulating and incorporating it into IHL. This paper aims to examine this by comparing the technological advancement of AI across countries and the geopolitical state of the countries implementing AI into warfare to understand both the timing and type of regulations. While looking at these factors, this paper holds three expectations: governments will be more reactive than proactive when it comes to regulating AI in warfare. The

next expectation is that governments in war will react and impose regulations differently compared to governments in peacetime. Finally, governments in peacetime will still impose regulations reactively to keep ahead of their global competitors in a sort of arms race.

To understand this relationship between AI regulations and government, four different studies were looked at to understand the extent of how governments regulate AI, how quickly these implementations and regulations are put into place, the factors that cause these regulations and regulatory timings, and the differences between governments. These four studies were split up into two sections: types of governmental regulation across the world and types of governmental regulations across geopolitical states. The first group of papers was chosen to understand how governments were implementing and regulating AI both in general and for warfare. The next two papers were chosen to compare the extent and types of regulations based on if the country was in wartime compared to if a country was in peacetime.

Global Views on AI Regulation

The first group of papers that this paper looks at is the global views on AI regulations. The first paper is “Global views on AI regulation: Comparative Global AI Regulation: Policy Perspectives from the EU, China, and the US (Chun et al).” The paper looks at and compares how the EU, the United States of America, and China are approaching AI regulations in general, with some mentions of warfare. The paper also looked at which different factors influence these regulations. These questions were based on political structure, country economy, certain values and priorities, and competition among governments. The paper held the notions that political structures shape the way AI is regulated, there will be a trade-off based on the way that governments regulate AI,

and that the way global entities regulate AI is not driven by a single nation, but rather that it is a global effort. The paper does this by qualitatively analyzing various policies and frameworks publicly provided by the EU, the US, and China.

The paper found that all three groups regulated AI differently. The EU regulated AI in a more proactive manner when it came to commercial AI. They wanted to make sure that AI was being used responsibly and ethically, which caused innovation to be slower compared to the other two groups. However, despite this stance towards AI in the commercial sector, it did not include these safety regulations in the context of warfare. In this aspect, regulations were still more reactive than proactive. While looking at China, the paper found that there were a significant number of regulations. Through these regulations, however, they did not promote safety solely, but also promoted fast innovation in the development of AI. Their stance on regulation in the commercial sector was fairly similar to their regulations regarding military and defense. This will be further expanded upon later in this paper. The final group that they looked at was the US. They found that the US was a lot more hands-off in its regulation of AI. They relied a lot on pre-existing laws and regulations and were more market-driven. Based on this market drive, they primarily let the markets self-regulate in the commercial sector. Similarly to China, this stance in the commercial was echoed in their defence. This will also be expanded upon further later in this paper. After looking at these three examples, the paper explains that these policies are often not stagnant but develop around other countries' policies in both cooperation and competition with each other.

The next paper in this group is “Military AI Needs Technically-Informed Regulation to Safeguard AI Research and Its Applications (Simmons-Edler et al).” The question that the paper sought to answer was how existing frameworks in regulation fail to adequately address technical risks, and what can be done to patch these holes in AI regulations. This was done by looking at how AI is being used in warfare, the level of autonomy that is given to the AI, the regulations and standards prior to the use of AI, and the level of researcher involvement in the creation of these regulations. This is done to understand the performance of both the AI and how well the regulations hold up, the geopolitical stability within the regions using AI, and the overall effectiveness of the regulations. The paper also holds the notion that current IHL regulations do not perfectly encapsulate the use of AI in warfare, as the technology introduces new factors and discrepancies. They also have the expectation that AI researchers need to be involved in the process of creating regulations in order to create adequate military policies.

Similar to the previous paper, Simmons-Edler et al. look into their research question by qualitatively looking into policies from a wide variety of countries. They didn’t do this comparatively but focused on looking for vulnerabilities across all of the policies in three sectors: military, international relations, and institutional impacts. Through this research, the paper found that existing frameworks do not appropriately address the needs that come with AI governance. This conclusion came from a lack of proper definitions for important terms, such as “meaningful human control.” They also found that regulations are inadequate, as the current regulations regarding AI are mostly extensions of pre-existing laws, not newer regulations. Due to this, they found that there are no metrics when it comes to unique AI issues and that there are no apparent ways to solve them based on current laws. These issues become more apparent as

they also found, across various instances, HOTL military AI performed unpredictably and would not perform well outside of their initial testing scenarios. Based on all of this, the paper concludes by calling for five recommendations when creating regulations: a formal binding prohibition on AI control, international standards for AI laws, a ban on AI commanding over humans, clarification on regulations under IHL, and the inclusion of AI researchers when implementing regulations and military weaponry.

Based on these two papers, governments are primarily reactive in regulating AI in warfare. This includes countries with proactive policies and regulations regarding AI in the commercial sector. On top of this, the extent of the IHL based on current regulations is not sufficient to cover the new discrepancies that come with HOTL AI systems in warfare, as most of them are either not adequate or are laws made for humans, while not being updated for AI systems. This issue is further expanded upon. The papers suggested and highly advocated for including AI researchers, as it would make sure that laws and regulations stay informed, up to date with the latest developments in AI, and relevant to the contexts in which AI is being used.

Comparing Regulations Across Geopolitical States

The next group of papers focuses more on how countries in different geopolitical states formulate or adhere to regulations regarding AI in weaponry. The first countries that this paper looks at are Ukraine and Russia, as they are currently in conflict with each other. This case study was chosen as this war is recent enough to still be relevant, while enough time has passed since the start of the war that there is relevant research to come out of it. The paper that is being examined is “The Use of Autonomous Weapons in the Ukraine Conflict: Assessing Compliance

with IHL Principles (Khalid).” The paper seeks to answer the question of whether existing IHL frameworks are enough for the regulation of Autonomous Weapons Systems (AWS) in conflict and how the Ukraine-Russia conflict is responding. They do this with the expectation that Automated Weapons Systems used by Ukraine do not follow IHL, and existing regulations do not effectively work in places of combat due to the development of technology outpacing laws. They do this by qualitatively looking at various policy papers and case studies by comparing the autonomy of the AI systems, the contexts of use, and the capabilities of the systems to understand the compliance with IHL regulations, how clear the regulations are before they are broken, and the outcomes of the case studies after AI weaponry has been used.

Khalid looked through the different legal documents and relevant IHL frameworks in order to qualitatively analyze three case studies covering Ukrainian and Russian use of AI weaponry. They found that automated weapon systems were not consistent enough for HOTL AI systems to safely avoid civilian casualties and that human oversight was a necessary component when using them. Khalid worried that if HOTL systems were being used, it would cause a “responsibility black hole,” where decision-making is so spread out that it is hard or impossible to attribute IHL violations. Khalid also found that Ukraine would not follow all of the IHL regulations, as they prioritized their country's survival over following the IHL properly.

The next paper looks at two countries that are not in a wartime state: China and the United States. As explained in the first paper of this section, the development of AI regulations also comes about from competition between two countries. The paper, “‘Autonomous weapons’ as a geopolitical signifier in a national power play: analysing AI imaginaries in Chinese and US

military policies (Bächle et al),” focuses on how China and the US influence the regulations of each nation due to the competition between the two powers. The paper asks the question of how the US and China approach the concept of AWS in their militaries and in what ways do their approaches reflect their geopolitical power. It holds the expectations that China and the US will have ambiguous AWS definitions to develop weaponry while still being within regulations, and that international regulation is near impossible due to competition. They do this by looking at government interests in AWS, and geopolitical identity to examine how each nation categorizes AWS, how they negotiate in CCW, and progress towards international regulation.

Bächle et al. look at frameworks, publicly available UN position papers for Certain Conventional Weapons (CCW), and other publicly available national documents that showcase each country's position and regulations on AI weaponry. They focused on how each military's policies and their technical definitions for AWS. Bächle et al. found that the US had a significantly low bar for their definition of autonomy. The US also had vague frameworks, which made it hard to derive meaningful regulations. In contrast, China set the bar for autonomy incredibly high, which gives the guise of strict regulations. However, this allows China to create sophisticated AI weaponry while “not being good enough to be considered autonomous.” Both of these countries submitted CCW positions that were ambiguous, which allowed them to leave options open for AWS development. Both the ambiguity and the strictness of regulations and positions are a part of the strategies of these countries in order to be competitive with one another in the AWS arms race.

From these two papers, it is understood that the geopolitical state of a country is important when trying to understand how governments handle regulation. Countries that are in wartime are more

likely to be reactive when making regulations, less likely to have regulations, and less likely to properly follow the IHL, as they are prioritizing their survival. In contrast, countries that are in peacetime are more likely to have regulations that they attempt to follow. However, these regulations can vary, and often give governments a lot of space to be able to create the AI weaponry that they want to create. These loosened regulations are a product of the competition that governments have with each other.

The Economics of AI in Defense

To examine the ethics of AI in defense, it's important to not overlook the economic lens of its implementation and integration. During the postwar era (post WWII), the heavy defense R&D spending has been justified on two grounds: direct military utility and the broader civilian spillover that it generates. To properly examine the economic consequences of this spending, the following terms must be defined clearly:

- 1) Military Utility: the efficacy of the tool and its combative/strategic benefits for the military itself.
- 2) Civilian Spillover: the broader benefits of the innovation to the public.

Through these two justifications, the immense R&D spending in the U.S. has stayed theoretically justified throughout this era. The civilian population bore the cost of weapons development and received innovation in return.

This paper argues that AI represents a structural break from this model. Because AI's core inputs are produced in the civilian sector, defense AI investment no longer generates the broad spillovers that have historically supplemented its justification. The economic case for defense AI

spending is therefore almost entirely dependent on military utility, rather than in conjunction with the tangible benefits from civilian spillovers. This shift substantively alters the ethical calculus that the following sections address.

To evaluate this claim, this section of the paper first examines the historical justification to evaluate its validity. Then, it will identify a weakening of the traditional spillover model. Lastly, it will explore real-world evidence of AI's reversal of this structure and the consequences of this new dynamic.

The Historical Justification: Defense R&D as a Source of Civilian Spillover

To address the reversal of the traditional spillover model, it's important to first verify that defense R&D has historically generated the civilian spillovers it has been credited with producing. In the first paper, "The Intellectual Spoils of War? Defense R&D, Productivity and International Spillovers", Moretti, Steinwender, and Van Reenen provide the empirical foundation of this structure's existence.

By using OECD panel data from 1987-2009, they find a 10% increase in government-funded defense R&D leads to around a 5% increase in privately funded R&D. This is a crowding-in effect, an effect which promotes growth in the private sector positively with defense R&D. In the US aerospace industry alone in 2002, \$3 billion in defense R&D produced an estimated \$1.9 billion in additional private R&D.

As the basis of this paper's argument, it is important to understand how the authors arrived at this claim, especially considering the possibility of confounding variables in such a broad and complex area as industry growth. To isolate this causal effect, the authors employed an instrumental variable strategy. They utilized the predicted defense R&D in conjunction with the historical country industry share. By comparing the actual budget shocks with the predicted budget, they were able to see the effect on the typical industry share of the highlighted country. This allowed them to find the difference between the expected and actual budgets and the expected v.s. actual industry shares.

While the relationship is clear as a result of the econometric analysis performed by the authors, how was this spillover actually occurring? They categorized the spillover into the following three channels:

Direct Technology Transfer

- The classic examples of defense innovations that have revolutionized public life are members of this channel. The internet, GPS, and jet aviation all were born in defense R&D programs and are key figureheads for the civilian spillover justification model.

Crowd-in of Private R&D

- Defense contracts reduce fixed costs and financial barriers that limit private firms from pursuing R&D projects, making investments possible that would not otherwise occur.

Increase in trained technical workforce

- The necessary training for the technical roles involved with the technological innovation carries over from defense sector roles into the private markets. The human capital is invested in and remains valuable across the R&D landscape.

The spillover that the authors identified was not just an economic externality, they were the foundational element of the political and ethical calculus that has sustained high defense spending for decades. Taxpayers fund defense spending, and namely defense R&D, with the understanding that there would be resulting innovation beyond military utility, this critical concept of civilian spillover. This calculus defines the dual benefits of defense R&D, and must be questioned if one of these benefits were to waver.

The Weakening of the Spillover Model

The civilian spillover model, while historically robust, did not hold uniformly across all sectors or across time. The postwar spillover model was showing signs of weakness before AI even entered the picture. Mowery's "Military R&D and Innovation" addresses this structural weakening through the lens of technology lifecycles and different technological sectors.

This paper found that in information technology, the parental precursor of AI, there was already a reversal by 2010. Civilian innovation outpaced and drove military capability, rather than the other way around. This reversal is exemplified by the semiconductor sector. The Department of Defense actually invested \$900 million in the Very High Speed Integrated Circuit program in an attempt to restore defense leadership in chips, and the program failed because the commercial markets had well surpassed the capabilities of what defense R&D or procurement could reach.

The mechanism behind this reversal is simple. As a technology matures, civilian markets grow larger than their military counterparts. Commercial firms are able to move faster and invest more heavily, aiming to profit from broader market incentives. Military requirements remain narrower

and more specialized, diverging from the innovation in the civilian markets. This mechanism results in defense R&D losing its position on the frontier and becoming a follower rather than an innovator.

This pattern is critical to understanding what AI represents. The IT reversal Mowery demonstrates was sector-specific, and was a precursor to AI, with core inputs being civilian-produced, and the technical knowledge required for innovation demanding and scarce. Information technology was the first indication of a reversal of this direction of innovation flow, and AI has inherited this reversal as its starting position.

AI as a Complete Inversion

Williams and Evans' "The Defence Economics of Artificial Intelligence and Machine Learning" identify why AI cannot follow the historical flow of innovation in principle. The AI industry functions as an enabling technology with four core inputs: data, compute, algorithms, and model architectures. All four of these inputs are produced primarily in the civilian sector, and are improving at a rapid rate. The military is forced to adopt AI and follow civilian development, rather than lead it. This is not a temporary structure to allow defense R&D to gain a foothold on the frontier, but a permanent structural shift, akin to the IT reversal, this time from conception. The scale at which training data is produced, the open source research culture through which architectures progress, and the commercial cloud infrastructure that provides the necessary computer are all features of the civilian markets that defense R&D cannot replicate internally.

The contemporary evidence confirms this claim. In July 2025, the Department of Defense awarded \$200 million in AI contracts to Anthropic, Google, OpenAI, and xAI, the frontier private AI firms. Not a single traditional defense firm received a contract for R&D. This reveals the current DoD preferences and capabilities: they cannot produce frontier AI capabilities internally, and must purchase it from the commercial sector.

The February 2026 standoff with Anthropic reinforces this point and demonstrates the power shift when the traditional spillover model reverses. The Pentagon demanded that Anthropic remove safeguards barring government autonomous weapons and mass domestic surveillance from their technology, and Anthropic refused. The Trump administration, in turn, threatened to invoke the Defense Production Act, revealing their dependence on Anthropic. Anthropic refused to alter their guidelines, setting a new precedent in AI where commercial firms set the terms of military use of their own technology, a precedent possible only when this spillover model flows from commercial to defense.

Conclusion

When it comes to the implementation of AWS, by combining a policy choice of limiting civilian casualties with advanced HITL systems, AI weapons could reduce collateral damage and unnecessary death in future conflicts. This reality has already started to be implemented, and further advances in AI weapons, when used with care and responsibility, can be used to better adhere to IHL.

When it comes to regulations in IHL, it is understood that the extent of the regulations is dependent on the individual countries, their relationship to one another, and the geopolitical state of the region. The timeliness of regulations across the board has been more reactive, as the development of technology outpaces the development of regulations. Finally, the factors that influence regulations depend on the geopolitical state. Countries at war will be less likely to prioritize following regulations, as they will be prioritizing survival, while countries out of war will be more focused on making sure their regulations fit their needs in competition.

For the economic side, the spillover model that historically made up half of the justification of defense R&D spending is structurally reversed in the AI context. Civilian AI is the driver of innovation and defense purchases from the frontier rather than creating. AI represents a fixed dependence and reversal of innovation flow. With the dual benefit argument no longer valid, the ethical justification for defense R&D in AI must be reevaluated, as taxpayers are now funding military utility solely, rather than receiving revolutionary civilian innovation as well.

The spillover model once allowed the government to point to technologies like the internet, GPS, and semiconductor industries as evidence for what defense R&D spending can produce for the public. With that argument gone for AI spending, the government must rely on a much narrower justification, military utility, and shift its ethical calculus accordingly.

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